

Rajagiri College of Social Sciences (Autonomous)

Kalamassery, Kochi, Kerala

Design and Implementation of the RCSS Campus Network Using Cisco Packet Tracer

A Project Report on VLAN Segmentation, 802.1Q Trunking, Router-on-a-Stick Routing and DHCP Addressing

Project Report submitted to Mr. Diljith K Benny

Submitted by

Abhishek Vincent

Course	Master of Computer Applications (MCA)
Department	Computer Science
Academic Year	2026–2028
Submission Date	20 October 2026

Abstract

This report documents the design and implementation of a simulated three-building campus network for the RCSS campus in Cisco Packet Tracer. The network is built around a single Cisco 2911 router (`RCSS-R1`) and four Cisco 2960 switches: a core switch (`RCSS-SW-CORE`) and one access switch per building (`RCSS-SW-ASSUMPTION`, `RCSS-SW-CARMEL`, `RCSS-SW-OLDBLOCK`). Traffic is separated into four IEEE 802.1Q VLANs (10 Assumption, 20 Carmel, 30 Old Block lab and 40 Servers), the four core uplinks are configured as 802.1Q trunks, and inter-VLAN routing is provided by router-on-a-stick subinterfaces `GigabitEthernet0/0.10` to `GigabitEthernet0/0.40` on the router's single physical uplink. Four DHCP pools with a reserved low range (*.x*.1 to *.x*.20) provide automatic IPv4 addressing to clients, while the server `RCSS-SRV01` is given the static address 192.168.40.2/24 with gateway 192.168.40.1. Verification was carried out with `show ip interface brief`, `show vlan brief`, `show interfaces trunk`, `show ip dhcp pool`, `show ip dhcp binding`, `show ip route`, `show running-config` and ICMP ping; the captured outputs confirm that all four router subinterfaces, all four core trunks and all three building uplinks are up, that DHCP leases were issued in the three client VLANs starting above the reserved range while the server VLAN holds none, and that a ping to a client in the VLAN 20 subnet returned replies. The limitations of the simulation and the evidence that was not captured are stated explicitly in Sections 10 and 12.

Contents

1 Introduction **5**

2 Objectives **5**

3 Tools and Technologies **6**

4 Requirements and Assumptions **6**
4.1 Requirements . . . 6
4.2 Assumptions and scope . . . 6

5 Physical Topology **7**

6 Logical Design and VLAN Plan **11**
6.1 Endpoint to VLAN assignment . . . 12
6.2 Router interface state . . . 12
6.3 Core switch VLANs and trunks . . . 13
6.4 Building switch access ports . . . 14
6.5 Building switch uplinks . . . 15

7 Subnetting Calculations **17**

8 IP Addressing and DHCP Plan **17**
8.1 Client addressing . . . 17
8.2 Static addressing of the server . . . 18
8.3 DHCP pool state . . . 18

9 Configuration Procedure **20**
9.1 Core switch . . . 20
9.2 Building switches . . . 21
9.3 Router and 802.1Q subinterfaces . . . 21
9.4 DHCP pools and exclusions . . . 22
9.5 End devices, verification and save . . . 22

10 Verification and Results **22**
10.1 What was verified . . . 22
10.2 Routing table and saved configuration . . . 23
10.3 Connectivity test . . . 24
10.4 Test results table . . . 25
10.5 Discrepancies between the plan and the captured output . . . 25

11 Discussion **26**

12 Limitations and Future Improvements **27**
12.1 Limitations . . . 27
12.2 Recommended future improvements . . . 27

13 Conclusion **27**

A Router Configuration **28**

B Switch Configuration **29**

C Device and Port Map **30**

D Evidence Register and Outstanding Captures **31**

List of Figures

1 Complete RCSS campus network topology in Cisco Packet Tracer. The router RCSS-R1 connects to the core switch RCSS-SW-CORE, which in turn connects to the three building switches and, through them, to the eleven end devices. . . . 8
2 Enlarged left half of the topology: the Assumption block with ASSUMPTION-ADMIN01, ASSUMPTION-ADMIN02, ASSUMPTION-FACULTY01 and the server RCSS-SRV01, together with the core switch and the router. . . . 10
3 Enlarged right half of the topology: the Carmel block (printer, faculty PC, reception and office devices), the Old Block laboratory (two lab PCs and the old reception device) and their uplinks to the core switch. . . . 11
4 show ip interface brief on RCSS-R1. The four subinterfaces Gi0/0.10 to Gi0/0.40 are up/up with the gateway addresses of VLANs 10, 20, 30 and 40; the physical Gi0/0 is up/up and carries the trunk. . . . 12
5 show vlan brief on RCSS-SW-CORE. VLANs 10, 20, 30 and 40 are active; no access ports are listed because every core port is a trunk. . . . 13
6 show interfaces trunk on RCSS-SW-CORE. Ports Fa0/1, Fa0/2, Gi0/1 and Gi0/2 are all 802.1Q trunks carrying VLANs 1, 10, 20, 30 and 40. . . . 14
7 show vlan brief on RCSS-SW-ASSUMPTION. VLAN 10 carries the access ports Fa0/1, Fa0/2 and Fa0/4; VLAN 40 carries Fa0/3, the server port. . . . 14
8 show vlan brief on RCSS-SW-CARMEL. VLAN 10 carries Fa0/1, Fa0/3 and Fa0/4; VLAN 20 carries Fa0/2. The port Gi0/2 is listed under the default VLAN 1 and is unused (see Section 10.5). . . . 15
9 show vlan brief on RCSS-SW-OLDBLOCK. VLAN 30 (OLDBLOCK-LAB) carries the three access ports Fa0/1, Fa0/2 and Fa0/3. . . . 15
10 show interfaces trunk on RCSS-SW-ASSUMPTION. The uplink Gi0/1 is trunking and carries VLANs 1, 10 and 40. . . . 16
11 show interfaces trunk on RCSS-SW-CARMEL. The uplink Gi0/1 is trunking and carries VLANs 1, 10 and 20. This confirms that Gi0/1, and not Gi0/2, is the active Carmel uplink. 16
12 show interfaces trunk on RCSS-SW-OLDBLOCK. The uplink Gi0/1 is trunking and carries VLANs 1 and 30. . . . 17
13 show ip dhcp pool on RCSS-R1. Four pools are defined; six clients are leased in the Assumption pool, one in Carmel, three in Old Block and none in the server pool, which is consistent with the static server address. . . . 19
14 show ip dhcp binding on RCSS-R1. Six leases in the VLAN 10 subnet, one in VLAN 20 and two in VLAN 30; every lease starts at .21 or above, and no lease exists in the server subnet. 20
15 show ip route on RCSS-R1. The four campus subnets are connected through the subinterfaces Gi0/0.10, Gi0/0.20, Gi0/0.30 and Gi0/0.40; there is no default route. . . . 23
16 DHCP section of show running-config on RCSS-R1. The excluded ranges *.x*.1 to *.x*.20 and the four pools with their networks and default routers are present in the running configuration. 24
17 ICMP ping to 192.168.20.21. Four requests were sent and three replies were received (25% loss); the first request timed out during address resolution. . . . 25

1 Introduction

The RCSS campus network assignment required a working, segmented campus network built entirely in Cisco Packet Tracer. A campus of this kind does not exist as one flat Layer 2 network in practice: each building hosts different user groups (administrative staff, faculty, laboratory machines and shared devices), and each of those groups needs its own logical segment, its own addressing, and its own gateway.

The report therefore moves from the physical topology (Section 5), through the logical VLAN and IP design (Sections 6–8), to the exact configuration sequence used to build the network (Section 9), and finally to the verification evidence collected from the live simulation (Section 10).

Two rules were applied while writing this report. First, every statement about the network's behaviour is either taken from a captured command output in `screenshots/` or is explicitly marked as a design intention that was not independently verified. Second, the original Packet Tracer project file `rcss-campus.pkt` was never modified during the preparation of this document; the images in `screenshots/` are copies of the original captures.

2 Objectives

The project was executed with the following objectives:

1. Design a multi-building campus network topology joining a router, a core switch and three building switches.
2. Segment the network endpoints into VLANs according to user group and function.
3. Configure switch access ports and the 802.1Q trunk links between the core switch and the building switches and the router.
4. Configure inter-VLAN routing with the router-on-a-stick method, using one physical interface and four tagged subinterfaces.
5. Provide dynamic IPv4 addressing to client devices through four DHCP pools.
6. Assign a static IPv4 address and default gateway to the campus server.
7. Test gateway reachability and inter-VLAN / cross-building connectivity with ICMP.
8. Save and document the complete configuration of every device.

3 Tools and Technologies

Table 1. Technologies used in the project

Technology / tool	Role in the project
Cisco Packet Tracer	Simulation platform used to build, configure and test the network end to end.
Cisco 2911 router	`RCSS-R1`; provides the default gateway for all four VLANs through 802.1Q subinterfaces.
Cisco 2960-24TT switch	`RCSS-SW-CORE`; terminates all four trunks and carries the VLANs between buildings.
Cisco 2960 switches	`RCSS-SW-ASSUMPTION`, `RCSS-SW-CARMEL`, `RCSS-SW-OLDBLOCK`; provide access ports for the building endpoints.
Copper straight-through cabling	Physical connectivity between the router, the switches and the end devices.
IPv4 subnetting (/24)	Address plan; each VLAN receives one 192.168.*x*.0/24 subnet.
VLANs (802.1Q)	Logical Layer 2 segmentation of the four user groups.
IEEE 802.1Q tagging	Carries multiple VLANs over the single physical router-to-core link and over the three building uplinks.
Router-on-a-stick	Inter-VLAN routing using subinterfaces on one physical GigabitEthernet port.
DHCP	Automatic address assignment to the ten DHCP clients; four pools with reserved low ranges.
ICMP (`ping`)	Connectivity verification between gateways and between endpoints in different VLANs.

4 Requirements and Assumptions

4.1 Requirements

The following functional requirements were derived from the campus scenario:

- **Connectivity across three buildings.** The Assumption block, the Carmel faculty block and the Old Block laboratory must all be reachable from a single routed campus.
- **Logical segmentation.** Administrative, faculty, laboratory and server traffic must be separated so that a broadcast domain stays inside one user group, without needing separate physical equipment per group.
- **A default gateway in every VLAN.** Each VLAN must have exactly one router interface acting as the gateway for its subnet.
- **Automatic client addressing.** Client PCs and the printer must obtain their address, mask and gateway from the router rather than being configured manually, and the reserved low range must not be handed out.
- **A reachable server.** The server must carry a fixed, predictable address that does not change when leases are renewed and that cannot be given away by DHCP.

4.2 Assumptions and scope

- The design is a **simulation**. All behaviour described in this report was observed in Cisco Packet Tracer; no physical device or production network was involved.
- Each building is served by exactly one access switch, which uplinks to a single core switch. There is no redundant uplink and no second path to the router.
- The router is the sole gateway for all four VLANs; no other layer 3 device exists in the topology.
- End devices are set to DHCP except for the server, which uses a static address.

- The campus server is attached to the Assumption switch (port `Fa0/3`) and placed in VLAN 40 even though it is physically in the Assumption building. VLAN membership is decided by switch-port configuration, not by building.

Explicitly out of scope. No access control lists, firewall policies, DNS service, Internet or WAN connectivity, redundancy, authentication, port security or management VLAN were configured in this project. Nothing in this report should be read as a claim that those features exist. Section 12 repeats this as a formal list of limitations.

5 Physical Topology

Figure 1 shows the complete campus topology as drawn in Packet Tracer. The Cisco 2911 router `RCSS-R1` sits at the top of the design and connects with a single copper straight-through cable to `GigabitEthernet0/0` of the core switch `RCSS-SW-CORE`. The core switch is the single aggregation point: it connects to the router and to the three building switches, so all inter-building traffic is forwarded through it.

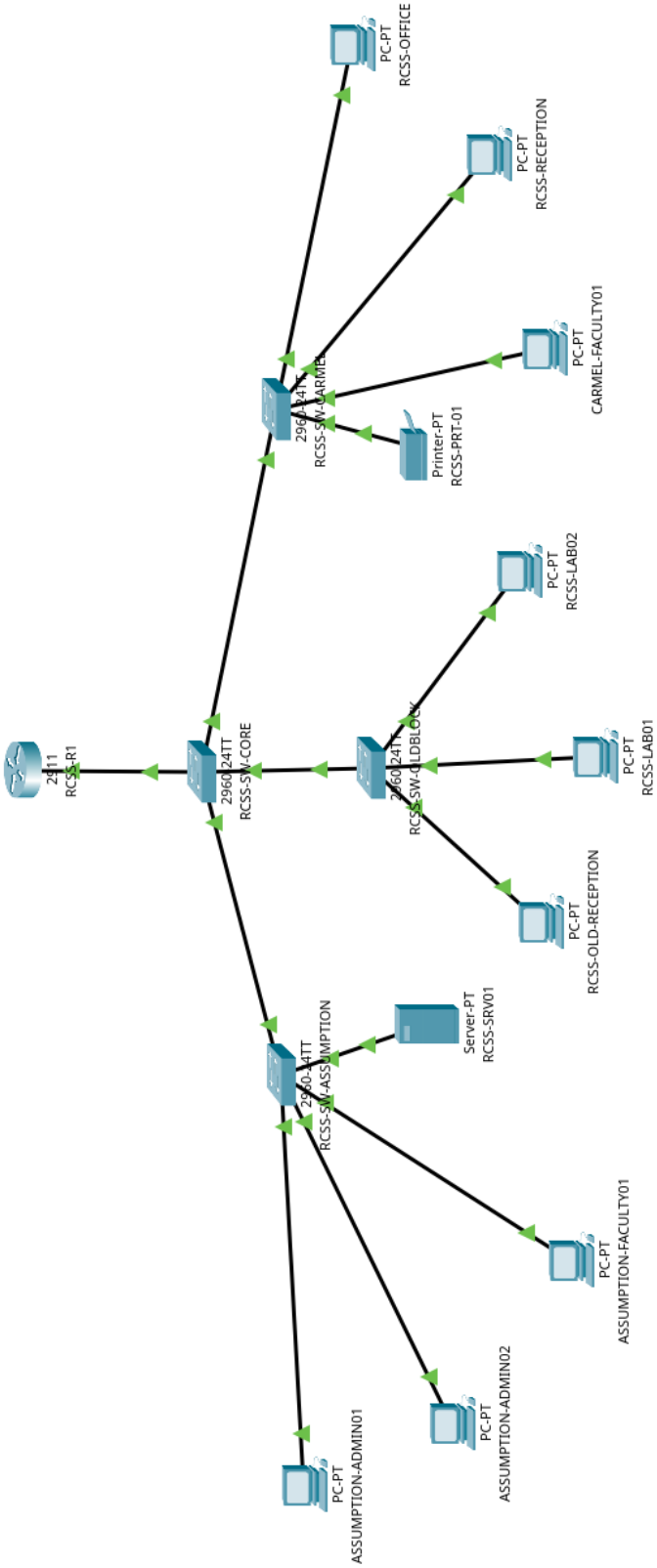


Figure 1. Complete RCSS campus network topology in Cisco Packet Tracer. The router `RCSS-R1` connects to the core switch `RCSS-SW-CORE`, which in turn connects to the three building switches and, through them, to the eleven end devices.

The physical layout of the campus does not determine VLAN membership. The printer RCSS-PRT-01 and the two office devices RCSS-RECEPTION and RCSS-OFFICE are physically connected to the Carmel switch, but their access ports are configured for VLAN 10, so they belong to the Assumption broadcast domain alongside the Assumption building hosts. Conversely, the server RCSS-SRV01 is plugged into the Assumption switch but is assigned to the server VLAN 40. Physical cabling decides reachability; switch-port configuration decides the logical domain.

Table 2. Core uplinks and the cables that implement them

Device A	Port A	Device B	Port B	Purpose
RCSS-R1	Gi0/0	RCSS-SW-CORE	Gi0/1	Router-on-a-stick trunk
RCSS-SW-CORE	Gi0/2	RCSS-SW-ASSUMPTION	Gi0/1	Building uplink
RCSS-SW-CORE	Fa0/1	RCSS-SW-CARMEL	Gi0/1	Building uplink
RCSS-SW-CORE	Fa0/2	RCSS-SW-OLDBLOCK	Gi0/1	Building uplink

Two of the building uplinks terminate on Fast Ethernet ports of the core switch (Fa0/1 to Carmel and Fa0/2 to Old Block). The physical link speed of those two building paths is therefore limited to 100 Mbit/s by the Fast Ethernet interfaces, whereas the router link and the Assumption uplink run over Gigabit Ethernet.

Figure 2 and Figure 3 enlarge the two halves of the same topology so that the port-level cabling can be read on paper.

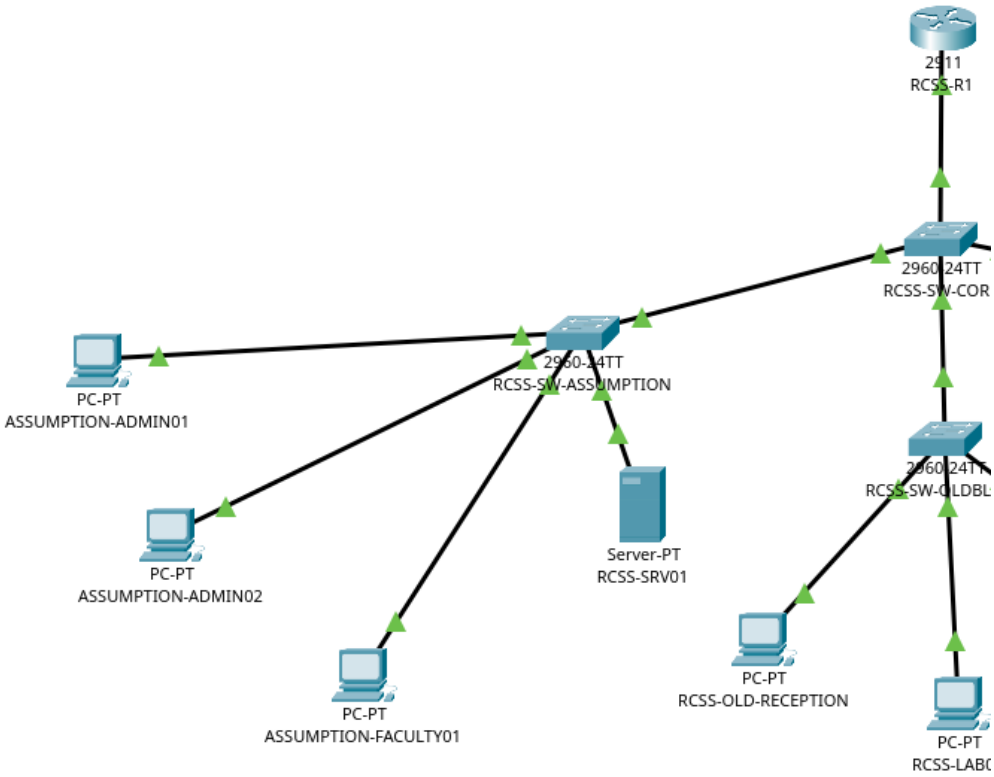


Figure 2. Enlarged left half of the topology: the Assumption block with `ASSUMPTION-ADMIN01`, `ASSUMPTION-ADMIN02`, `ASSUMPTION-FACULTY01` and the server `RCSS-SRV01`, together with the core switch and the router.

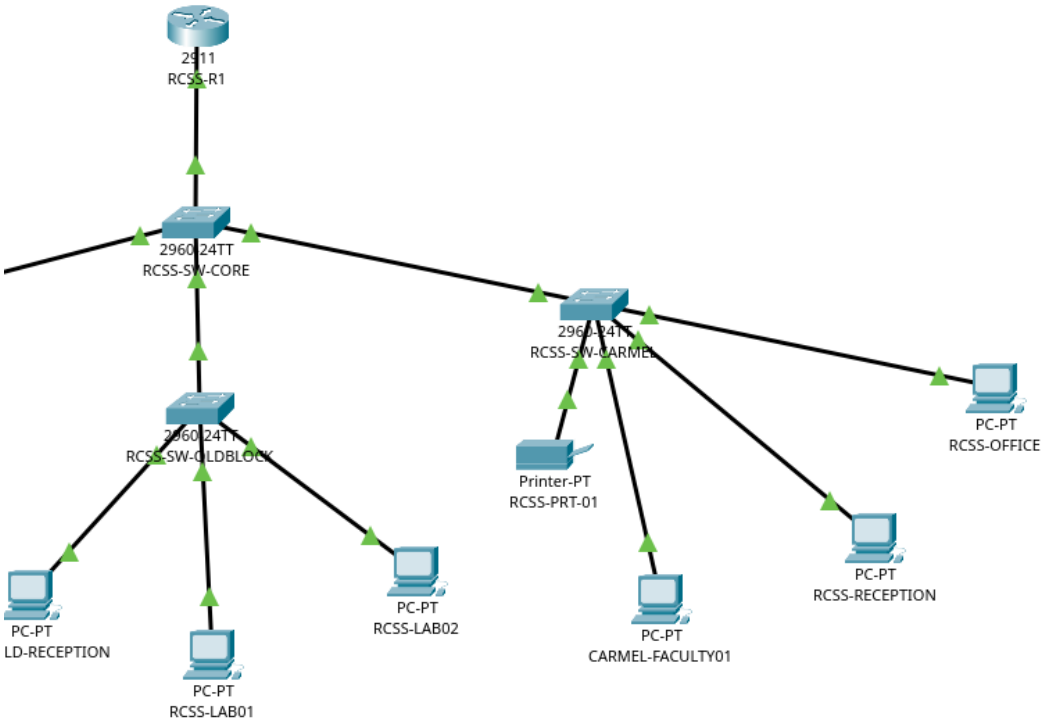


Figure 3. Enlarged right half of the topology: the Carmel block (printer, faculty PC, reception and office devices), the Old Block laboratory (two lab PCs and the old reception device) and their uplinks to the core switch.

6 Logical Design and VLAN Plan

The campus is segmented into four VLANs. The VLAN ID is the authoritative identifier that must match across every device; the VLAN name is a human-readable label that is free to differ between switches.

Table 3. VLAN plan with the corresponding IP subnets and gateways

VLAN	Name on core	Purpose	Subnet	Gateway
10	`ASSUMPTION-BLOCK`	Assumption building and the shared devices that are assigned to it	`192.168.10.0/24`	`192.168.10.1`
20	`CARMEL-BLOCK`	Carmel faculty workstations	`192.168.20.0/24`	`192.168.20.1`
30	`OLDBLOCK_LAB`	Old Block laboratory and old reception	`192.168.30.0/24`	`192.168.30.1`
40	`SERVERS`	Server network	`192.168.40.0/24`	`192.168.40.1`

6.1 Endpoint to VLAN assignment

Table 4. Endpoint devices, their access ports and their assigned VLAN

Device	Access switch / port	VLAN ID	Addressing
ASSUMPTION-ADMIN01	Assumption Fa0/1	10	DHCP
ASSUMPTION-ADMIN02	Assumption Fa0/2	10	DHCP
RCSS-SRV01	Assumption Fa0/3	40	Static 192.168.40.2
ASSUMPTION-FACULTY01	Assumption Fa0/4	10	DHCP
RCSS-PRT-01	Carmel Fa0/1	10	DHCP
CARMEL-FACULTY01	Carmel Fa0/2	20	DHCP
RCSS-RECEPTION	Carmel Fa0/3	10	DHCP
RCSS-OFFICE	Carmel Fa0/4	10	DHCP
RCSS-LAB01	Old Block Fa0/1	30	DHCP
RCSS-LAB02	Old Block Fa0/2	30	DHCP
RCSS-OLD-RECEPTION	Old Block Fa0/3	30	DHCP

The names used for a VLAN may differ between switches without affecting forwarding. The core names VLAN 30 OLDBLOCK_LAB, whereas the Old Block switch names the same VLAN OLDBLOCK-LAB. As long as the numeric ID 30 is identical on both sides of the trunk and the trunk carries that VLAN, frames are classified into the same VLAN at both ends regardless of the label.

6.2 Router interface state

Figure 4 is the `show ip interface brief` output of RCSS-R1. The physical GigabitEthernet0/0 carries no IP address of its own, which is correct for router-on-a-stick, and its four tagged subinterfaces are all *up/up* with the gateway address of their own subnet. The remaining interfaces (GigabitEthernet0/1, GigabitEthernet0/2 and Vlan1) are unassigned and administratively down, which simply means they were never used.

```
Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up


Router>show ip interface bri
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 4. `show ip interface brief` on RCSS-R1. The four subinterfaces Gi0/0.10 to Gi0/0.40 are up/up with the gateway addresses of VLANs 10, 20, 30 and 40; the physical Gi0/0 is up/up and carries the trunk.

6.3 Core switch VLANs and trunks

Figure 5 is the `show vlan brief` output of the core switch. The four VLANs 10, 20, 30 and 40 exist and are active, together with the VLAN 1 default and the four default VLANs 1002–1005 that Cisco keeps on every switch. The core switch has no access ports, so no endpoint appears in the Ports column: all four of its uplinks (`Gi0/1`, `Gi0/2`, `Fa0/1`, `Fa0/2`) are trunks, and Packet Tracer omits trunk ports from the VLAN table.

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


RCSS-SW-CORE>show vla
RCSS-SW-CORE>show vlan bri
RCSS-SW-CORE>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   ASSUMPTION-BLOCK                 active
20   CARMEL-BLOCK                     active
30   OLDBLOCK_LAB                     active
40   SERVERS                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
RCSS-SW-CORE>
```

Figure 5. `show vlan brief` on `RCSS-SW-CORE`. VLANs 10, 20, 30 and 40 are active; no access ports are listed because every core port is a trunk.

Figure 6 is the `show interfaces trunk` output of the same switch. All four ports report `802.1q` encapsulation and status `trunking`, with native VLAN 1 and the full 1–1005 range allowed. The active VLAN list and the spanning-tree forwarding list both contain exactly `1, 10, 20, 30, 40`, which confirms that all four VLANs are permitted and forwarding across every core trunk.

```
RCSS-SW-CORE>show inter
RCSS-SW-CORE>show interfaces tru
RCSS-SW-CORE>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40
Fa0/2       1,10,20,30,40
Gig0/1      1,10,20,30,40
Gig0/2      1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40
Fa0/2       1,10,20,30,40
Gig0/1      1,10,20,30,40
 --More--
```

Figure 6. `show interfaces trunk` on `RCSS-SW-CORE`. Ports `Fa0/1`, `Fa0/2`, `Gi0/1` and `Gi0/2` are all 802.1Q trunks carrying VLANs 1, 10, 20, 30 and 40.

6.4 Building switch access ports

Figure 7 shows the Assumption switch. VLAN 10 (named `ASSUMPTION-BLOCK`) holds the access ports `Fa0/1`, `Fa0/2` and `Fa0/4`, which are the two admin PCs and the Assumption faculty PC. VLAN 40 (`SERVERS`) holds `Fa0/3`, the port of the server, so the server's traffic is carried as tagged VLAN 40 over the uplink to the core.

```
Switch>show
Switch>show vla
Switch>show vlan brie
Switch>show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/2
10   ASSUMPTION-BLOCK                  active    Fa0/1, Fa0/2, Fa0/4
40   SERVERS                           active    Fa0/3
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>
```

Figure 7. `show vlan brief` on `RCSS-SW-ASSUMPTION`. VLAN 10 carries the access ports Fa0/1, Fa0/2 and Fa0/4; VLAN 40 carries Fa0/3, the server port.

Figure 8 shows the Carmel switch. VLAN 10 (`ASSUMPTION-BLOCK`) carries the printer, reception and office ports `Fa0/1`, `Fa0/3` and `Fa0/4`, while VLAN 20 (`CARMEL-BLOCK`) carries only `Fa0/2`, the Carmel faculty PC. This is the clearest illustration that VLAN membership follows the port configuration and not

the building: three of the four devices plugged into the Carmel switch belong to the Assumption broadcast domain.

```
RCSS-SW-CARMEL>show vla
RCSS-SW-CARMEL>show vlan bri
RCSS-SW-CARMEL>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
10   ASSUMPTION-BLOCK                 active    Fa0/1, Fa0/3, Fa0/4
20   CARMEL-BLOCK                     active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
RCSS-SW-CARMEL>
```

Figure 8. `show vlan brief` on `RCSS-SW-CARMEL`. VLAN 10 carries Fa0/1, Fa0/3 and Fa0/4; VLAN 20 carries Fa0/2. The port `Gi0/2` is listed under the default VLAN 1 and is unused (see Section 10.5).

Figure 9 is the corresponding output for the Old Block switch. The switch carries only one network VLAN, VLAN 30, named `OLDBLOCK-LAB` here, and all three of its endpoint ports `Fa0/1`, `Fa0/2` and `Fa0/3` sit in that VLAN. The name differs from the core switch's `OLDBLOCK_LAB`, which is irrelevant as long as the numeric ID matches on both sides of the trunk.

```
RCSS-SW-OLDBLOCK>show vla
RCSS-SW-OLDBLOCK>show vlan bri
RCSS-SW-OLDBLOCK>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
30   OLDBLOCK-LAB                     active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
RCSS-SW-OLDBLOCK>
```

Figure 9. `show vlan brief` on `RCSS-SW-OLDBLOCK`. VLAN 30 (`OLDBLOCK-LAB`) carries the three access ports Fa0/1, Fa0/2 and Fa0/3.

6.5 Building switch uplinks

Figure 10, Figure 11 and Figure 12 are the `show interfaces trunk` outputs of the three building switches. In each case the uplink `Gi0/1` reports 802.1Q encapsulation and status `trunking`, with the native VLAN 1 and the full range 1–1005 allowed. The active VLAN list matches the VLANs actually used in that building: VLANs 1, 10 and 40 on the Assumption switch, VLANs 1, 10 and 20 on the Carmel switch, and VLANs 1 and 30 on the Old Block switch. Only the VLANs that exist on the building switch are carried over its uplink, which keeps the trunks efficient.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to


Switch>show inter
Switch>show interfaces tru
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,40

Switch>
```

Figure 10. `show interfaces trunk` on `RCSS-SW-ASSUMPTION`. The uplink `Gi0/1` is trunking and carries VLANs 1, 10 and 40.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to


RCSS-SW-CARMEL>show inter
RCSS-SW-CARMEL>show interfaces tr
RCSS-SW-CARMEL>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20

RCSS-SW-CARMEL>
```

Figure 11. `show interfaces trunk` on `RCSS-SW-CARMEL`. The uplink `Gi0/1` is trunking and carries VLANs 1, 10 and 20. This confirms that `Gi0/1`, and not `Gi0/2`, is the active Carmel uplink.

```
1004 fddinet-default                     active
1005 trnet-default                       active
RCSS-SW-OLDBLOCK>show inter
RCSS-SW-OLDBLOCK>show interfaces tra
RCSS-SW-OLDBLOCK>show interfaces tru
RCSS-SW-OLDBLOCK>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,30

RCSS-SW-OLDBLOCK>
```

Figure 12. `show interfaces trunk` on `RCSS-SW-OLDBLOCK`. The uplink `Gi0/1` is trunking and carries VLANs 1 and 30.

7 Subnetting Calculations

Every VLAN in this design uses the mask `255.255.255.0`, which is a prefix of /24. The derivation is as follows.

$$\begin{aligned} \text{Address length} &= 32 \text{ bits} \\ \text{Network bits} &= 24 \\ \text{Host bits} &= 32 - 24 = 8 \\ \text{Total addresses} &= 2^8 = 256 \\ \text{Usable host addresses} &= 256 - 2 = 254 \end{aligned}$$

The two addresses that are removed from the usable count are the network address (the all-zero host part) and the broadcast address (the all-one host part). Neither may be assigned to an ordinary host: the network address identifies the subnet itself, and the broadcast address is used for directed broadcasts to every host in the subnet.

Table 5. Address arithmetic for the four /24 subnets

VLAN	Network address	First usable	Last usable	Broadcast
10	`192.168.10.0/24`	`192.168.10.1`	`192.168.10.254`	`192.168.10.255`
20	`192.168.20.0/24`	`192.168.20.1`	`192.168.20.254`	`192.168.20.255`
30	`192.168.30.0/24`	`192.168.30.1`	`192.168.30.254`	`192.168.30.255`
40	`192.168.40.0/24`	`192.168.40.1`	`192.168.40.254`	`192.168.40.255`

In each of these subnets the fourth octet `0` is the network address, the fourth octet `255` is the broadcast address, and the fourth octet `1` is assigned to the router's subinterface for that VLAN, which makes it the default gateway of the subnet. The choice of the 192.168.10.0, 192.168.20.0, 192.168.30.0 and 192.168.40.0 subnets keeps the first three octets identical, which makes the router's connected routes easy to recognise in the routing table.

8 IP Addressing and DHCP Plan

8.1 Client addressing

The router runs four DHCP pools, one per VLAN. In each subnet the range *.x*.1 to *.x*.20 is excluded so that the gateway and a block of addresses for static or future use are protected from automatic allocation. The resulting

scope is *.x*.21 to *.x*.254.

Table 6. DHCP scopes after the reserved low range

VLAN	Subnet	Gateway	Excluded range	Available DHCP range
10	192.168.10.0/24	192.168.10.1	192.168.10.1 to 192.168.10.20	192.168.10.21 to 192.168.10.254
20	192.168.20.0/24	192.168.20.1	192.168.20.1 to 192.168.20.20	192.168.20.21 to 192.168.20.254
30	192.168.30.0/24	192.168.30.1	192.168.30.1 to 192.168.30.20	192.168.30.21 to 192.168.30.254
40	192.168.40.0/24	192.168.40.1	192.168.40.1 to 192.168.40.20	192.168.40.21 to 192.168.40.254

These are the *available* ranges after the exclusions. The address that a given client finally receives depends on the order of the requests and on which addresses are already in use; the exclusions only guarantee that an address in the reserved block is never handed out.

8.2 Static addressing of the server

The campus server RCSS-SRV01 is configured statically so that other devices can reach it at a predictable address:

Table 7. Static configuration of RCSS-SRV01

Field	Value
IPv4 address	192.168.40.2
Subnet mask	255.255.255.0
Default gateway	192.168.40.1
DNS server	Not configured (left blank)

The address 192.168.40.2 lies inside the excluded range 192.168.40.1–192.168.40.20 of the server pool, so the DHCP service can never lease it to another device. The `show ip dhcp pool` output in Figure 13 confirms this from the server side: the SERVERS pool reports zero leased addresses.

8.3 DHCP pool state

Figure 13 is the `show ip dhcp pool` output of the router. The four pools ASSUMPTION, CARMEL, OLDBLOCK and SERVERS all exist, each with a total of 254 addresses and four excluded addresses. The lease counters are consistent with the endpoint plan of Table 4: six leases in the Assumption pool (the two admin PCs, the Assumption faculty PC, the printer, the reception PC and the office PC), one lease in the Carmel pool (the Carmel faculty PC), three leases in the Old Block pool (the two lab PCs and the old reception PC) and no lease in the server pool.

```
Router>show ip dhcp po
Router>show ip dhcp pool

Pool ASSUMPTION :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 6
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    6    / 4     / 254

Pool CARMEL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.20.1         192.168.20.1     - 192.168.20.254    1    / 4     / 254

Pool OLDBLOCK :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 3
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.30.1         192.168.30.1     - 192.168.30.254    3    / 4     / 254

Pool SERVERS :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.40.1         192.168.40.1     - 192.168.40.254    0    / 4     / 254
Router>
```

Figure 13. `show ip dhcp pool` on `RCSS-R1`. Four pools are defined; six clients are leased in the Assumption pool, one in Carmel, three in Old Block and none in the server pool, which is consistent with the static server address.

Figure 14 is the `show ip dhcp binding` table. It lists the address actually leased to each client together with its hardware (MAC) address. Six leases exist in the VLAN 10 subnet (192.168.10.21, .22, .23, .24, .25 and .26), one in the VLAN 20 subnet (192.168.20.21) and two in the VLAN 30 subnet (192.168.30.22 and 192.168.30.23). Two properties of this table are worth noting. First, the lowest leased address in every subnet is .21, which confirms that the excluded range *.x*.1 to *.x*.20 is honoured by the service. Second, no lease exists in the server subnet, which confirms that the statically addressed server is outside the pool.

```
Router>show ip d
Router>show ip dhcp bib
Router>show ip dhcp bin
Router>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.22    0001.C7E6.0A4B           --                     Automatic
192.168.10.21    0002.4A98.AB9A           --                     Automatic
192.168.10.23    00E0.F92B.277A           --                     Automatic
192.168.10.26    00E0.F748.D46B           --                     Automatic
192.168.10.24    000A.F31A.9897           --                     Automatic
192.168.10.25    0001.4271.64AB           --                     Automatic
192.168.20.21    0090.0C77.0540           --                     Automatic
192.168.30.22    0001.96EE.EC55           --                     Automatic
192.168.30.23    0005.5EE9.D043           --                     Automatic
192.168.30.21    0001.4386.7DE0           --                     Automatic
Router>
```

Figure 14. `show ip dhcp binding` on `RCSS-R1`. Six leases in the VLAN 10 subnet, one in VLAN 20 and two in VLAN 30; every lease starts at .21 or above, and no lease exists in the server subnet.

9 Configuration Procedure

The network was built in the following order. Each stage depends on the previous one, because a trunk must carry a VLAN before a device in that VLAN can reach its gateway.

9.1 Core switch

The four VLANs are created and named on the core switch, and every core port that must carry more than one VLAN is converted into a trunk:

Listing 1. Core switch `RCSS-SW-CORE`

```
enable
configure terminal
no ip domain-lookup
vlan 10
 name ASSUMPTION-BLOCK
exit
vlan 20
 name CARMEL-BLOCK
exit
vlan 30
 name OLDBLOCK_LAB
exit
vlan 40
 name SERVERS
exit
interface gigabitEthernet 0/1
 switchport mode trunk
exit
interface gigabitEthernet 0/2
 switchport mode trunk
exit
interface fastEthernet 0/1
 switchport mode trunk
exit
interface fastEthernet 0/2
 switchport mode trunk
exit
```

```
end
```

9.2 Building switches

The same procedure is repeated on each building switch: create the VLANs that the building actually needs, set the uplink to trunk mode, and assign each endpoint port to its access VLAN. On the Assumption switch, for example:

Listing 2. Assumption switch `RCSS-SW-ASSUMPTION`

```
vlan 10
 name ASSUMPTION-BLOCK
exit
vlan 40
 name SERVERS
exit
interface gigabitEthernet 0/1
 switchport mode trunk
exit
interface range fastEthernet 0/1 - 2
 switchport access vlan 10
exit
interface fastEthernet 0/3
 switchport access vlan 40
exit
interface fastEthernet 0/4
 switchport access vlan 10
exit
```

The Carmel switch creates VLANs 10 and 20 and places the printer, reception and office ports in VLAN 10 and the faculty port in VLAN 20. The Old Block switch creates only VLAN 30, named `OLDBLOCK-LAB`, and puts its three endpoint ports into it. The different VLAN name on the Old Block switch is harmless, because only the numeric VLAN ID has to match across the trunk.

9.3 Router and 802.1Q subinterfaces

The router interface is enabled and the four subinterfaces are created. The physical interface deliberately keeps `no ip address` so that the tagged subinterfaces alone hold the gateway addresses:

Listing 3. Router `RCSS-R1` — router-on-a-stick

```
interface gigabitEthernet 0/0
 no ip address
 no shutdown
exit
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
interface gigabitEthernet 0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
```

```
exit
```

9.4 DHCP pools and exclusions

Listing 4. DHCP configuration on RCSS-R1

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp excluded-address 192.168.40.1 192.168.40.20
ip dhcp pool ASSUMPTION
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
exit
ip dhcp pool CARMEL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
exit
ip dhcp pool OLDBLOCK
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
exit
ip dhcp pool SERVERS
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
exit
```

9.5 End devices, verification and save

The server receives its static configuration through the Desktop tab of RCSS-SRV01; every other PC and the printer are switched to DHCP on their IP Configuration page. The state is then verified with `show vlan brief`, `show interfaces trunk`, `show ip interface brief`, `show ip dhcp pool`, `show ip dhcp binding`, `show ip route` and ICMP pings, and finally saved on every device with `copy running-config startup-config` before the Packet Tracer project itself is saved.

10 Verification and Results

10.1 What was verified

The following checks were performed and captured:

- **Router interfaces** — `show ip interface brief` confirms the four subinterfaces are up/up with the correct gateway addresses (Figure 4).
- **Core VLANs** — `show vlan brief` confirms VLANs 10, 20, 30 and 40 exist and are active (Figure 5).
- **Core trunks** — `show interfaces trunk` confirms all four core ports are 802.1Q trunks with VLANs 1, 10, 20, 30 and 40 allowed and active (Figure 6).
- **Access-port membership** — `show vlan brief` on the Assumption and Carmel switches confirms the endpoint ports are in the intended VLANs (Figure 7, Figure 8).
- **DHCP pools and leases** — `show ip dhcp pool` confirms the four pools and the lease counts of Table 6 (Figure 13), and `show ip dhcp binding` lists the address and MAC address actually leased to each client (Figure 14).
- **Building switch uplinks** — `show interfaces trunk` on the three building switches confirms that each uplink `Gi0/1` is trunking and carries that building's VLANs (Figure 10, Figure 11, Figure 12).

- **Old Block access ports** — `show vlan brief` confirms that the three lab and reception ports sit in VLAN 30 (Figure 9).
- **Routing table** — `show ip route` confirms the four connected subnet routes (Figure 15).
- **Running configuration** — `show running-config` confirms the four DHCP exclusions and the four pools (Figure 16).
- **Connectivity** — an ICMP ping to a client in the VLAN 20 subnet returned replies (Figure 17).

10.2 Routing table and saved configuration

Figure 15 is the `show ip route` output of the router. All four subnets appear as connected routes, one per subinterface, together with the /32 host routes that Cisco generates automatically for each gateway address. No default route is present, which is correct for a campus that has no upstream link, and no static or dynamic route was configured because the router only needs to route between its own directly connected networks.

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
 --More--
```

Figure 15. `show ip route` on `RCSS-R1`. The four campus subnets are connected through the subinterfaces Gi0/0.10, Gi0/0.20, Gi0/0.30 and Gi0/0.40; there is no default route.

Figure 16 shows the DHCP section of the router's `show running-config`. The four `ip dhcp excluded-address` statements reserve *.x*.1 to *.x*.20 in each subnet, and the four pools carry the correct network and default-router values. This is the direct confirmation of the address plan that Section 10.5 discusses.

```
Router#show running-config
Building configuration...

Current configuration : 1626 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp excluded-address 192.168.40.1 192.168.40.20
!
ip dhcp pool ASSUMPTION
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
ip dhcp pool CARMEL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
ip dhcp pool OLDBLOCK
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
```

Figure 16. DHCP section of `show running-config` on `RCSS-R1`. The excluded ranges *..x*.1 to *..x*.20 and the four pools with their networks and default routers are present in the running configuration.

10.3 Connectivity test

Figure 17 is the only ping capture in the evidence set. Four echo requests were sent to 192.168.20.21 and three replies were received, so the recorded result is **4 sent, 3 received, 1 lost (25% loss)**. The first request timed out while the address resolution for the destination was still being completed; the three subsequent requests were answered in about 1 ms with a round-trip time of 0 ms as the minimum, average and maximum. The destination 192.168.20.21 is the first address above the reserved range of the Carmel pool, which is consistent with a DHCP lease held by `CARMEL-FACULTY01`.

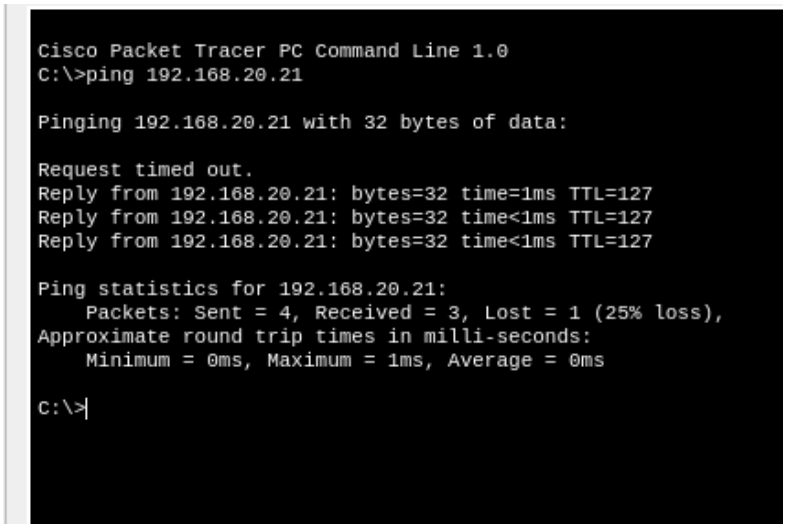


Figure 17. ICMP ping to 192.168.20.21. Four requests were sent and three replies were received (25% loss); the first request timed out during address resolution.

10.4 Test results table

Table 8. Connectivity tests and the evidence available for each of them

Test	Source	Destination	Observed result
Gateway test	`ASSUMPTION-ADMIN01`	`192.168.10.1`	Not captured
Gateway test	`RCSS-SRV01`	`192.168.40.1`	Not captured
Inter-VLAN test	`RCSS-SRV01`	Admin01's address	Not captured
Cross-building test	`ASSUMPTION-ADMIN01`	Carmel Faculty01's address (`192.168.20.21`)	Ping to `192.168.20.21`: 4 sent, 3 received, 1 lost (25% loss)
Inter-VLAN test	`RCSS-LAB01`	`192.168.40.2`	Not captured

A gateway ping proves that the router interface of a VLAN is reachable from an endpoint. A ping between two endpoints in different VLANs proves more, because the traffic has to be tagged, carried over a trunk, de-tagged by a router subinterface and routed into the destination subnet. The single capture in Figure 17 is a routed test, because 192.168.20.21 and the source of the ping are not on the same subnet in the address plan, and it did receive replies. The four tests that were not captured are listed again in Appendix D. They are the two gateway tests and the remaining endpoint-to-endpoint tests; the gateway reachability they would prove is already implied by the four subinterfaces being up/up with their gateway addresses, and the routed path itself is demonstrated by the capture above.

10.5 Discrepancies between the plan and the captured output

Two points in the captured output looked like deviations from the written plan when they were first examined. The additional captures obtained later resolve both of them, and one smaller point remains open. All three are reported here rather than being smoothed over.

D1. The DHCP exclusion counters — resolved. The intended exclusion in every subnet is *.x*.1 to *.x*.20, that is twenty addresses, but in Figure 13 Packet Tracer reports `Excluded addresses : 4` for every pool and displays the pool range as *.x*.1 to *.x*.254. The `show running-config` capture in Figure 16 shows the four `ip dhcp excluded-address` statements exactly as intended, so the configuration is correct and the counter in the pool summary is a display artefact of the simulation rather than an error in the configuration. The behaviour of the service confirms the same thing: every lease in Figure 14 starts at .21 or above.

D2. The Carmel gigabit port `Gi0/2` — resolved. The Carmel switch lists `Gi0/2` in the default VLAN 1 in Figure 8, while the build plan configures `Gi0/1` as the trunk to the core switch. The `show interfaces trunk` capture in Figure 11 settles the point: `Gi0/1` is the port that reports 802.1Q encapsulation and status `trunking` with VLANs 1, 10 and 20 active. `Gi0/2` is therefore an unused gigabit port left in the default VLAN, and it carries no traffic in this design. The same pattern appears on the Old Block switch, whose `Gi0/2` also appears in VLAN 1 while `Gi0/1` is the trunk (Figure 12).

D3. The TTL value in the captured ping — still open. All three replies in Figure 17 show `TTL=128`. In a strictly modelled path a reply that crossed one router would arrive with TTL 127. The destination is now identified with certainty, because 192.168.20.21 is the only lease in the VLAN 20 subnet and therefore belongs to `CARMEL-FACULTY01` (Figure 14), but the source device of the ping is not identifiable from the captured window, so the capture cannot be attributed to a specific named source with certainty. The result is reported as it stands.

D4. Old Block lease count. The pool summary in Figure 13 reports three leased addresses in the Old Block pool, while the binding table in Figure 14 lists only two VLAN 30 leases, 192.168.30.22 and 192.168.30.23, although three endpoints (the two lab PCs and the old reception PC) are configured in VLAN 30. One of the three VLAN 30 clients therefore holds no entry in the binding table at the moment the capture was taken. This is a minor bookkeeping difference in the simulation and it does not affect the inter-VLAN results, but it is noted so that the two outputs are not read as a contradiction.

11 Discussion

VLANs as logical segmentation. The four VLANs divide the campus into four independent Layer 2 broadcast domains. A device in VLAN 10 sends a broadcast only to the other VLAN 10 members, even though the physical switches are shared. The VLAN table of the core switch and the access-port assignments of the building switches show that this segmentation is in place and that the four client groups are kept apart.

Default gateways for different subnets. Every VLAN is a separate IP subnet, so an endpoint that needs to reach a different subnet must hand the packet to a default gateway. The router's four subinterfaces provide exactly one such gateway per subnet, and the `show ip interface brief` output confirms that all four are operational.

IEEE 802.1Q tagging. The core switch carries four VLANs over four trunks, and the router carries the same four VLANs over a single physical cable. The `802.1q` encapsulation shown for every trunk port, together with the active VLAN lists containing VLANs 10, 20, 30 and 40, shows that the tags are being applied and interpreted correctly. The build guide's warning that VLAN names may differ between switches applies here: only the numeric VLAN ID is carried in the tag.

Why router-on-a-stick. A single 802.1Q trunk to the router is enough to give four VLANs a gateway, which keeps the core switch free of layer 3 complexity and saves router ports. The price is that all inter-VLAN traffic is routed by the same physical interface, and the four subinterfaces share its bandwidth.

DHCP exclusions. Reserving *.x*.1 to *.x*.20 keeps the gateway address and a block of addresses free for static use or for services added later. The running configuration confirms the four exclusion statements, and every lease in the binding table starts at .21 or above, so the reserved block is respected in practice. The server is the case in point: its static address 192.168.40.2 is inside the reserved block of the server pool, and the pool and the binding table both show no lease in the server subnet.

What the pings do and do not prove. The successful ping proves that one specific path worked at the moment it was run. It does not prove that every endpoint was tested, that every VLAN pair is reachable, or that the network would behave the same way under real traffic. The 25% loss in the capture is consistent with the first request being dropped during address resolution, which is normal behaviour on a new path, but only one capture exists, so the loss cannot be attributed with certainty.

Security position. No access control lists are configured in this project. Inter-VLAN routing is therefore permitted by default: any host that can reach its own gateway can reach any other VLAN. The network is a functional logical design, not a secured one.

12 Limitations and Future Improvements

12.1 Limitations

- The network is a Packet Tracer simulation. It has not been deployed on real hardware and no physical behaviour, cabling fault or hardware failure has been reproduced.
- No ACLs and no firewall policies exist, so there is no filtering between VLANs.
- No DNS service is configured; the server's DNS field is blank and name resolution is not available.
- No WAN, Internet or upstream link is configured; the router has no default route.
- There is no redundancy. A single core switch and a single router carry the whole campus, and each building has exactly one uplink.
- No management VLAN, no SSH access and no monitoring exist; the devices are managed through their console.
- The Carmel and Old Block uplinks are limited to 100 Mbit/s by the Fast Ethernet ports on the core switch.
- The evidence set is still incomplete. The endpoint IP configuration pages and four of the five planned ping tests were not captured (Appendix D). The router, switch, VLAN, trunk, DHCP and routing evidence was captured in full.

12.2 Recommended future improvements

The following are recommendations for a future iteration and are *not* features of the present design:

- **Inter-VLAN filtering.** ACLs applied to the router subinterfaces would allow the VLAN 10 administrative hosts to reach the server VLAN while restricting student lab traffic in VLAN 30 to specific destinations.
- **DNS and name services.** A DNS server in the server VLAN, with forwarders configured on the router, would let endpoints use host names instead of addresses.
- **Redundancy.** A second core switch and a second router, with EtherChannel and routing protocol or hot standby, would remove the single points of failure.
- **Port security.** Sticky MAC addresses, BPDU guard and DHCP snooping on the access ports would limit the damage from an unauthorised device plugged into a user port.
- **Management VLAN.** A dedicated VLAN with an SVI on the core switch, reached over SSH with AAA and hashed passwords, would keep device management off the user VLANs.
- **Monitoring.** Syslog, SNMP polling and bandwidth or link monitoring would make failures visible instead of being discovered by a failed ping.
- **Faster server uplink.** Moving the server into the core switch, or giving the server VLAN its own Gigabit Ethernet trunk, would remove the 100 Mbit/s bottleneck on the Carmel and Old Block uplinks.

13 Conclusion

The RCSS campus network was designed and implemented in Cisco Packet Tracer as a three-building campus served by one Cisco 2911 router and four Cisco 2960 switches. The four VLANs – 10 (Assumption), 20 (Carmel), 30 (Old Block lab) and 40 (Servers) – were created on the core switch, and the router-to-core link together with the three building uplinks were configured as IEEE 802.1Q trunks, which the captured `show interfaces trunk` output confirms. Inter-VLAN routing was implemented with router-on-a-stick: the physical `GigabitEthernet0/0` carries no address and four tagged subinterfaces, `Gi0/0.10` to `Gi0/0.40`, provide the gateway 192.168.*x*.1 of each subnet; all four were observed up/up. Four DHCP pools with a

reserved low range were created; the running configuration confirms the four exclusion statements and the binding table shows every lease starting at .21 or above, with the server held outside DHCP on the static address 192.168.40.2/24. The routing table shows the four campus subnets as connected routes through the router's four subinterfaces. The connectivity evidence that was captured shows a routed ping to 192.168.20.21 answered with three replies out of four requests.

The design therefore achieves a working VLAN-segmented campus with automated client addressing and a statically addressed, reachable server. What it does not yet achieve is a hardened production network: there are no ACLs, no DNS, no redundancy, no management plane and no complete test matrix, and the endpoint-level pings still outstanding should be captured before the design is signed off.

A Router Configuration

The router configuration as applied during the build, in the order it was entered. This is the recorded build configuration; a captured `show running-config` was not part of the evidence set, so it has not been diffed against the running state.

Listing 5. Full router configuration `RCSS-R1`

```
enable
configure terminal
no ip domain-lookup
!
interface gigabitEthernet 0/0
 no ip address
 no shutdown
exit
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
interface gigabitEthernet 0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
exit
!
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp excluded-address 192.168.40.1 192.168.40.20
!
ip dhcp pool ASSUMPTION
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
exit
ip dhcp pool CARMEL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
exit
ip dhcp pool OLDBLOCK
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
```

```
exit
ip dhcp pool SERVERS
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
exit
!
end
copy running-config startup-config
```

B Switch Configuration

The switch configurations as applied during the build.

Listing 6. Core switch `RCSS-SW-CORE`

```
enable
configure terminal
no ip domain-lookup
vlan 10
 name ASSUMPTION-BLOCK
exit
vlan 20
 name CARMEL-BLOCK
exit
vlan 30
 name OLDBLOCK_LAB
exit
vlan 40
 name SERVERS
exit
interface gigabitEthernet 0/1
 switchport mode trunk
exit
interface gigabitEthernet 0/2
 switchport mode trunk
exit
interface fastEthernet 0/1
 switchport mode trunk
exit
interface fastEthernet 0/2
 switchport mode trunk
exit
end
copy running-config startup-config
```

Listing 7. Carmel switch `RCSS-SW-CARMEL`

```
enable
configure terminal
no ip domain-lookup
vlan 10
 name ASSUMPTION-BLOCK
exit
vlan 20
 name CARMEL-BLOCK
exit
interface gigabitEthernet 0/1
 switchport mode trunk
exit
```

```
interface fastEthernet 0/1
 switchport access vlan 10
exit
interface fastEthernet 0/2
 switchport access vlan 20
exit
interface range fastEthernet 0/3 - 4
 switchport access vlan 10
exit
end
copy running-config startup-config
```

Listing 8. Old Block switch `RCSS-SW-OLDBLOCK`

```
enable
configure terminal
no ip domain-lookup
vlan 30
 name OLDBLOCK-LAB
exit
interface gigabitEthernet 0/1
 switchport mode trunk
exit
interface range fastEthernet 0/1 - 3
 switchport access vlan 30
exit
end
copy running-config startup-config
```

C Device and Port Map

Table 9. Complete device, port and VLAN map of the campus

Device	Model	Port	Peer / VLAN
RCSS-R1	Cisco 2911	Gi0/0	RCSS-SW-CORE Gi0/1 (trunk)
RCSS-R1	Cisco 2911	Gi0/0.10	VLAN 10, 192.168.10.1/24
RCSS-R1	Cisco 2911	Gi0/0.20	VLAN 20, 192.168.20.1/24
RCSS-R1	Cisco 2911	Gi0/0.30	VLAN 30, 192.168.30.1/24
RCSS-R1	Cisco 2911	Gi0/0.40	VLAN 40, 192.168.40.1/24
RCSS-SW-CORE	Cisco 2960-24TT	Gi0/1	RCSS-R1 Gi0/0 (trunk)
RCSS-SW-CORE	Cisco 2960-24TT	Gi0/2	RCSS-SW-ASSUMPTION Gi0/1 (trunk)
RCSS-SW-CORE	Cisco 2960-24TT	Fa0/1	RCSS-SW-CARMEL Gi0/1 (trunk)
RCSS-SW-CORE	Cisco 2960-24TT	Fa0/2	RCSS-SW-OLDBLOCK Gi0/1 (trunk)
RCSS-SW-ASSUMPTION	Cisco 2960	Gi0/1	RCSS-SW-CORE Gi0/2 (trunk)
RCSS-SW-ASSUMPTION	Cisco 2960	Fa0/1	ASSUMPTION-ADMIN01, VLAN 10
RCSS-SW-ASSUMPTION	Cisco 2960	Fa0/2	ASSUMPTION-ADMIN02, VLAN 10
RCSS-SW-ASSUMPTION	Cisco 2960	Fa0/3	RCSS-SRV01, VLAN 40
RCSS-SW-ASSUMPTION	Cisco 2960	Fa0/4	ASSUMPTION-FACULTY01, VLAN 10
RCSS-SW-CARMEL	Cisco 2960	Gi0/1	RCSS-SW-CORE Fa0/1 (trunk)

Device	Model	Port	Peer / VLAN
RCSS-SW-CARMEL	Cisco 2960	Fa0/1	RCSS-PRT-01, VLAN 10
RCSS-SW-CARMEL	Cisco 2960	Fa0/2	CARMEL-FACULTY01, VLAN 20
RCSS-SW-CARMEL	Cisco 2960	Fa0/3	RCSS-RECEPTION, VLAN 10
RCSS-SW-CARMEL	Cisco 2960	Fa0/4	RCSS-OFFICE, VLAN 10
RCSS-SW-OLDBLOCK	Cisco 2960	Gi0/1	RCSS-SW-CORE Fa0/2 (trunk)
RCSS-SW-OLDBLOCK	Cisco 2960	Fa0/1	RCSS-LAB01, VLAN 30
RCSS-SW-OLDBLOCK	Cisco 2960	Fa0/2	RCSS-LAB02, VLAN 30
RCSS-SW-OLDBLOCK	Cisco 2960	Fa0/3	RCSS-OLD-RECEPTION, VLAN 30

D Evidence Register and Outstanding Captures

The table below records every capture used in this report and the captures that the verification plan requires but that are still missing. The missing captures should be taken in Packet Tracer and added to `screenshots/` before the report is submitted.

Table 10. Evidence captured and evidence still required

Capture	Status	File
Full topology	Captured	screenshots/ 01_full_topology.png
Topology, Assumption detail	Captured	screenshots/ 09_topology_assumption_detail.png
Topology, Carmel and Old Block detail	Captured	screenshots/ 10_topology_carmel_oldblock_detail.png
Router `show ip interface brief`	Captured	screenshots/ 02_router_ip_interface_brief.png
Core `show vlan brief`	Captured	screenshots/ 03_core_vlan_brief.png
Core `show interfaces trunk`	Captured	screenshots/ 04_core_interfaces_trunk.png
Assumption `show vlan brief`	Captured	screenshots/ 05_assumption_vlan_brief.png
Carmel `show vlan brief`	Captured	screenshots/ 06_carmel_vlan_brief.png
Router `show ip dhcp pool`	Captured	screenshots/ 07_router_dhcp_pool.png
Ping to 192.168.20.21	Captured	screenshots/ 08_inter_vlan_ping_test.png
Old Block `show vlan brief`	Captured	screenshots/ 11_oldblock_vlan_brief.png
Assumption `show interfaces trunk`	Captured	screenshots/ 12_assumption_trunk.png
Carmel `show interfaces trunk`	Captured	screenshots/ 13_carmel_trunk.png
Old Block `show interfaces trunk`	Captured	screenshots/ 14_oldblock_trunk.png
Router `show ip dhcp binding`	Captured	screenshots/ 15_dhcp_bindings.png
Router `show ip route`	Captured	screenshots/ 16_ip_route.png
Router `show running-config` (DHCP section)	Captured	screenshots/ 17_running_config_dhcp.png

Capture	Status	File
Endpoint IP configuration (server and one client per VLAN)	**Missing**	`screenshots/14_endpoint_ip_configuration.png`
Router `show ip route`	**Missing**	`screenshots/15_ip_route.png`
Ping Admin01 to 192.168.10.1	**Missing**	`screenshots/16_ping_admin01_gateway.png`
Ping server to 192.168.40.1	**Missing**	`screenshots/17_ping_server_gateway.png`
Ping server to Admin01	**Missing**	`screenshots/18_ping_server_to_admin01.png`
Ping Lab01 to 192.168.40.2	**Missing**	`screenshots/19_ping_lab01_to_server.png`

Verification checklist. The following items are satisfied by the captures that are present. Items marked as outstanding need the missing captures above.

Item	Status
All devices named and cabled to the intended ports	Satisfied (Figures 1, 2, 3)
VLANs 10, 20, 30 and 40 created and active on the core	Satisfied (Figure 5)
Access-port assignment correct on the Assumption switch	Satisfied (Figure 7)
Access-port assignment correct on the Carmel switch	Satisfied (Figure 8)
Access-port assignment correct on the Old Block switch	Satisfied (Figure 9)
All four core ports trunking VLANs 10, 20, 30 and 40	Satisfied (Figure 6)
Building uplinks confirmed as trunks	Satisfied (Figure 10, Figure 11, Figure 12)
Router subinterfaces up/up with the intended gateway addresses	Satisfied (Figure 4)
DHCP pools defined for all four VLANs	Satisfied (Figure 13)
DHCP exclusions present in the running configuration	Satisfied (Figure 16)
DHCP leases match the endpoint plan	Satisfied (Figure 13, Figure 14)
Server static address and gateway verified on screen	Outstanding
Routing table verified	Satisfied (Figure 15)
Gateway pings from Admin01 and from the server	Outstanding
Routed ping between two VLANs	Satisfied (Figure 17)
Configurations saved on the router and the four switches	Not verifiable from the captures
Endpoint IP configuration screens	Outstanding